

BCIS Subscription Billing and Collection System

Acceptance Test Report

Generated October 8, 2026 at 12:06 AM from the automated test output. Node v24.13.0, Windows 10.0.26200, PostgreSQL test database.

Total tests	Passed	Failed	Skipped
67	67	0	0

1. Mandatory acceptance tests (section 7 of the laboratory activity)

ID	Scenario	Expected result	Evidence	Result
AT-01	Exact payment (invoice 999.00, payment 999.00)	Remaining 0.00; invoice PAID; ledger balanced; receipt created.	2 automated tests	PASSED
AT-02	Partial payment (invoice 999.00, payment 500.00)	Remaining 499.00; PARTIALLY_PAID; allocation and ledger correct.	2 automated tests	PASSED
AT-03	Advance payment (monthly 1,000.00, payment 3,000.00)	Excess kept as advance credit and applied to later invoices; no value lost.	2 automated tests	PASSED
AT-04	Oldest-first arrears (Aug 999.00 + Sep 999.00, payment 1,200.00)	August 0.00; September 798.00 remaining.	3 automated tests	PASSED
AT-05	Duplicate GCash reference	Flagged at submission; verification and posting blocked.	2 automated tests	PASSED
AT-06	Payment reversal	Original visible; linked reversal; balances restored; actor and reason audited.	3 automated tests	PASSED
AT-07	Collector balanced remittance (20,000.00 / 20,000.00)	Difference 0.00; batch reconciles and closes.	2 automated tests	PASSED
AT-08	Collector shortage (20,000.00 / 19,500.00)	500.00 shortage displayed; cannot be closed as balanced.	3 automated tests	PASSED
AT-09	Concurrent users (three clients)	No corruption, duplicate numbering or cross-session effect.	1 automated test	PASSED
AT-10	Authorization (cashier calls admin-only operations)	Server rejects the operation even when the API is called directly.	5 automated tests	PASSED
AT-11	Duplicate billing (generation run twice)	No duplicate finalized invoice per service account and period.	2 automated tests	PASSED
AT-12	Backup and restore	Restored database passes the integrity check; expected records return.	2 automated tests	PASSED

Each scenario is exercised through the real HTTP API with authentication, authorization, validation and a real PostgreSQL database. The tests are in tests/acceptance/ and the domain rules are additionally unit tested in source/shared.

2. All automated tests

API acceptance · AT-01 Exact payment (1/1 passed)

invoice 999.00 + payment 999.00 -> remaining 0, PAID, ledger balanced, receipt created	1056 ms	PASSED
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API acceptance · AT-02 Partial payment (1/1 passed)

invoice 999.00 + payment 500.00 -> remaining 499.00, PARTIALLY_PAID	257 ms	PASSED
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API acceptance · AT-03 Advance payment (1/1 passed)

monthly 1,000.00 + payment 3,000.00 -> credit is kept and applied to future invoices without losing value	278 ms	PASSED
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API acceptance · AT-04 Oldest-first arrears (2/2 passed)

August 999.00 + September 999.00, payment 1,200.00 -> August 0.00, September 798.00	136 ms	PASSED
manual allocation needs authorization and cannot over-apply	129 ms	PASSED

API acceptance · AT-05 Duplicate GCash reference (2/2 passed)

a reference that backs a posted payment is flagged at submission and blocked at verification	248 ms	PASSED
rejects proof files that are not real images and never trusts the client file name	66 ms	PASSED

API acceptance · AT-06 Payment reversal (3/3 passed)

original stays visible, linked reversal exists, balances restore, actor and reason are audited	520 ms	PASSED
the database itself refuses to delete or edit posted financial records	199 ms	PASSED

a retried request with the same idempotency key posts only once	427 ms	PASSED
API acceptance · AT-07 Collector balanced remittance (1/1 passed)		
cash collected 20,000.00, remitted 20,000.00 -> difference 0.00; batch reconciles and closes	1257 ms	PASSED
API acceptance · AT-08 Collector shortage (2/2 passed)		
cash collected 20,000.00, remitted 19,500.00 -> 500.00 shortage shown; cannot be closed as balanced	740 ms	PASSED
overage is recorded the same way, and non-cash collections are not expected in the cash remittance	542 ms	PASSED
API acceptance · AT-09 Concurrent users (1/1 passed)		
three clients post and read at the same time: no corruption, no duplicate numbers, no cross-session effect	2768 ms	PASSED
API acceptance · AT-10 Authorization (5/5 passed)		
a cashier calling admin-only user and backup operations directly is rejected by the server	472 ms	PASSED
requests without a valid session are rejected on every protected route	5 ms	PASSED
every role receives exactly the permissions of the documented role matrix	1119 ms	PASSED
read-only and operational roles cannot mutate financial data	125 ms	PASSED
locks the account after repeated failed logins and supports session lock/unlock	1716 ms	PASSED
API acceptance · AT-11 Duplicate billing (2/2 passed)		
running billing generation twice for the same period creates no duplicate invoice	191 ms	PASSED
drafts are reviewed before finalizing, and a voided invoice can be re-billed once	204 ms	PASSED
API acceptance · AT-12 Backup and restore (2/2 passed)		
create backup, change data, restore the verified backup -> integrity check passes and expected records return	4331 ms	PASSED
refuses to restore a backup whose file was tampered with	804 ms	PASSED
API acceptance · billing rules (3/3 passed)		
plan price changes affect future billing only; historical invoices keep the billed rate	743 ms	PASSED
adjustments change a finalized invoice only through an audited, ledger-posted record	703 ms	PASSED
validation errors name the offending fields	79 ms	PASSED
API acceptance · receivables and service control (3/3 passed)		
aging totals reconcile to the outstanding invoice balances	363 ms	PASSED
suspension candidates follow the grace period and threshold; suspension and reconnection keep full history	447 ms	PASSED
subscribers and service accounts are closed by status, never deleted	68 ms	PASSED
API acceptance · search and reports (3/3 passed)		
global search finds subscribers, invoices, receipts and GCash references	113 ms	PASSED
every report builds, totals are consistent, and PDF/XLSX/CSV exports are real files	669 ms	PASSED
the dashboard KPIs agree with the receivables summary	157 ms	PASSED
Unit · money (integer centavos) (3/3 passed)		
parses user input without floating point	3 ms	PASSED
rejects invalid amounts	1 ms	PASSED
formats centavos for display	1 ms	PASSED
Unit · dates and billing periods (2/2 passed)		
handles month arithmetic across years	1 ms	PASSED
computes due dates and day differences	21 ms	PASSED
Unit · payment allocation - oldest unpaid invoice first (7/7 passed)		
AT-01 exact payment clears the invoice	2 ms	PASSED
AT-02 partial payment leaves the remainder on the invoice	0 ms	PASSED
AT-03 advance payment keeps the excess as unapplied credit - nothing is lost	0 ms	PASSED
AT-04 pays August before September regardless of input order	1 ms	PASSED
skips settled invoices and handles a subscriber with nothing due	1 ms	PASSED

rejects zero, negative and fractional amounts	0 ms	PASSED
always conserves the payment across many invoice mixes	15 ms	PASSED
Unit · manual allocation (authorized override) (2/2 passed)		
applies to the chosen invoice and keeps the rest as credit	0 ms	PASSED
refuses over-allocation, duplicates and foreign invoices	1 ms	PASSED
Unit · invoice totals and state machine (3/3 passed)		
adds charges and subtracts discounts	0 ms	PASSED
derives every status from amounts and dates	7 ms	PASSED
never allows a negative balance	0 ms	PASSED
Unit · accounts receivable aging (2/2 passed)		
uses the documented bucket boundaries	0 ms	PASSED
bucket totals reconcile to the outstanding balances	0 ms	PASSED
Unit · collector remittance reconciliation (3/3 passed)		
AT-07 balanced remittance	0 ms	PASSED
AT-08 shortage is reported, never rounded away	0 ms	PASSED
overage is reported	0 ms	PASSED
Unit · role and permission matrix (3/3 passed)		
gives the owner every permission and nobody an unknown one	3 ms	PASSED
keeps sensitive operations away from lower roles	1 ms	PASSED
read-only roles hold no mutating permission	0 ms	PASSED
Desktop end-to-end (Electron) (10/10 passed)		
the renderer is sandboxed: no Node.js, only the typed bridge	554 ms	PASSED
wrong password is rejected with a readable message	450 ms	PASSED
administrator: dashboard, subscriber profile, ledger and statement of account	4410 ms	PASSED
administrator: billing, invoices, receivables and aging	3934 ms	PASSED
administrator: GCash verification with duplicate-reference protection	1329 ms	PASSED
administrator: collection batch, remittance shortage and reconciliation	2395 ms	PASSED
administrator: reports, export to PDF and XLSX, audit trail	2602 ms	PASSED
cashier: receive a partial cash payment, preview the allocation, print the receipt	3362 ms	PASSED
cashier: role restrictions are enforced by the server, not only by hidden menus	848 ms	PASSED
owner: payment reversal with audit trail, session lock, backup	5943 ms	PASSED

3. How to reproduce

- `npm run db:start` (local PostgreSQL)
- `npm test` (unit + API acceptance tests against the `bcis_test` database)
- `npm run db:reset -- --yes && npm run build -w @bcis/desktop && npm run test:e2e` (desktop end-to-end tests and screenshots)
- `npm run test:report` (runs everything and regenerates this document)